Hayatullah Ahmadzai

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Website | Google Scholar

Education

2015–2019: Ph.D. in Economics (focus: Agricultural and Development Economics)
University of Nottingham, Nottingham, UK

2008–2011: M.S. in Agricultural Economics
Purdue University, West Lafayette, IN, USA

2002–2005: B.S. in Agronomy and Soil Sciences
Kabul University, Kabul, Afghanistan

Primary Research Interests: Agricultural & Food Policy • Climate Shocks, Adaptation, & Agricultural Resilience • Farm & Household Economics • Food Security and Rural Livelihoods • Applied Econometrics & Causal Inference

Academic and Research Appointments

2025–present: Research Fellowships and Affiliations

MESA Global Academy Scholar, Middle East Studies Association (MESA), Washington, DC, USA (remote; based in Durham, NC, USA). Awarded for 2025–2026 and competitively renewed for 2026–2027.
External Research Fellow (remote, unpaid), Centre for Research in Economic Development and International Trade (CREDIT), University of Nottingham, Nottingham, UK. Continuing affiliation following Research Fellow appointment (2022–2025).

06/2022–06/2025: Research Fellow
Centre for Research in Economic Development and International Trade (CREDIT), School of Economics, University of Nottingham, Nottingham, UK.

04/2019–04/2022: Postdoctoral Research Fellow
International Center for Biosaline Agriculture, Academic City, Dubai, UAE.

02/2012–09/2015: Senior Economist/Head of Monitoring, Evaluation & Learning (MEL)
Department for International Development (DfID, now FCDO), Kabul, Afghanistan.

03/2011–09/2015: Assistant Professor (Teaching-focused)
Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan *(held concurrently with the FLAG role, 2011–2012, and the DfID role, 2012–2015; 1–2 courses per term)*

07/2011–02/2012: Economist and Value-Chain Development Specialist
FLAG International Afghanistan (FIA), Kabul, Afghanistan.

03/2006–10/2007: Teaching Assistant
Department of Agronomy and Soil Sciences, Kabul University, Kabul, Afghanistan.

Publications

Peer-reviewed Journal Articles

Tests for market failures and overcoming transaction costs barriers to market participation in Afghanistan
Ahmadzai, H. 2026. *Journal of Social and Economic Development*, 28, 637–655.

Climate shocks, household food security and welfare in Afghanistan
Ahmadzai, H., and Morrissey, O. 2025. *Food Policy*, 134, 102910.

Crop Diversification, Household Welfare and Conflict: Afghanistan 2011–2017
Ahmadzai, H., and Morrissey, O. 2025. *The Journal of Development Studies*, 61(3), 381–399.

Empirical Analysis of Impacts of Sand and Dust Storms on Agricultural Productivity in Mongolia
Ahmadzai, H., Malhotra, A., and Tutundjian, S. 2023. *PLOS ONE*, 18(2), e0269271.

The Impact of Sand and Dust Storms on Agriculture Productivity in Iraq
Ahmadzai, H. 2023. *Middle East Development Journal*, 15(1), 50–65.

Hope for Change: Is Diversifying Production Portfolios an Ideal Strategy to Boost Farming Efficiency in Afghanistan?
Ahmadzai, H. 2022. *Progress in Development Studies*, 22(1), 7-31.

Orphan Crops: A Best Fit for Dietary Enrichment and Diversification in Highly Deteriorated Marginal Environments
Talabi, A., Vikram, P., Thushar, S., Hifzurrahman, R., Ahmadzai, H., Nhamo, N., Shahid, M., and Singh, R. 2022. *Frontiers in Plant Science*, 13, 839704.

Policies for Sustainable Agriculture and Livelihood in Marginal Lands: A Review
Ahmadzai, H., Elouafi, I., and Tutundjian, S. 2021. *Sustainability*, 13(16), 8692.

Development of Quinoa Value Chain to Improve Food and Nutritional Security in Rural Communities in Rehamna, Morocco: Lessons Learned and Perspectives
Hirich, A., Rafik, S., Rahmani, M., Fetouab, A., Filali, K., Ahmadzai, H., et al. 2021. *Plants*, 10(2), 301.

How is Off-farm Income Linked to On-farm Diversification? Evidence from Afghanistan
Ahmadzai, H. 2020. *Studies in Agricultural Economics*, 122(1), 1–12.

Trends in Quinoa Adoption in Marginal Areas, Economic Viability, and Policy Outlook
Ahmadzai, H. 2020. *Journal of Agribusiness and Rural Development*, 57(3), 235–247.

Book Chapters

Marginal Areas and Indigenous People Priorities for Research and Action
Sayed, A., and Ahmadzai, H., et al. 2023. In *Science and Innovations for Food Systems Transformation*, edited by von Braun, J., et al.,
pp. 261–279. Springer International Publishing.
(Expanded from the 2021 Food Systems Summit brief; see Policy Papers and Technical Reports.)

Impacts of Sand and Dust Storms on Agricultural Production
Middleton, N., Ahmadzai, H., and Ayurzana, E. 2023. In *Sand and Dust Storms: A Guide to Mitigation, Adaptation, Policy and Risk Management Measures in Agriculture*, edited by Middleton, N., pp. 33–49. FAO.

From Desert Farm to Fork: Value Chain Development for Innovative Salicornia-based Food Products in the United Arab Emirates
Lyra, D., Lampakis, E., …, Ahmadzai, H., et al. 2021. In *Future of Sustainable Agriculture in Saline Environments*, edited by Negacz, K., et al.,
pp. 181–199. CRC Press/Taylor & Francis Group.

Policy Papers and Technical Reports

Marginal Lands: Potential for Agricultural Development, Food Security and Poverty Reduction
Ahmadzai, H., Tutundjian, S., et al. 2022. *SOLAW21 Technical Background Report*, Rome: FAO.

Agri-tech: A Catalyst for Treated Wastewater Deployment in High Efficiency Food Production Systems
Ammar, K., Fatnassi, H., and Ahmadzai, H., et al. 2022. *International Center for Biosaline Agriculture (ICBA)*, Dubai, UAE.

Marginal Areas and Indigenous People: Priorities for Research and Action
Sayed, A., and Ahmadzai, H., et al. 2022. *Food Systems Summit Brief*. Food Systems Summit 2021.

Addressing Sand and Dust Storms in Northeast Asia (China, Mongolia), Western Asia (Iran, Iraq, and Kuwait), and North Africa (Algeria): An Agricultural Perspective
Ahmadzai, H. 2021. *Food and Agriculture Organization (FAO) and International Center for Biosaline Agriculture (ICBA).*

Baseline Survey Report: Improving Agricultural Resilience to Salinity (RESADE)
Ahmadzai, H. 2020. International Center for Biosaline Agriculture (ICBA), IFAD, BADEA.

Resarch in Progress

Security, Infrastructure, and Gender Gap in Rural Labor Markets: Evidence from Afghanistan's Repeated Cross-Sectional Surveys
Ahmadzai, H. 2026.
CREDIT Research Paper; Work in Progress

Research Projects and Funding

2022: IIE-SRF Afghanistan Emergency Initiative Fund, Institute of International Education-Scholar Rescue Fund (IIE-SRF), Cara (Council for At-Risk Academics), and the University of Nottingham. **Research Fellowship Award** (research conducted at CREDIT). Research support: £86,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Catalyzing Investments and Actions to Enhance Resilience Against Sand and Dust Storms in Agriculture. **Principal Investigator**. Grant Amount: $150,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Climate-Resilient and Gender-Inclusive Cropping Systems in Climate-Vulnerable Areas in Asia and the Pacific. **Co-Principal Investigator** with R.K. Singh (ICBA). Total Project Budget: $178,441.

2021: Ministry of Climate Change and Environment, UAE, Developing National Policy to Utilize Treated Sewage Effluent (TSE) and Biosolids in Controlled Environment Agriculture in the UAE. **Co-Principal Investigator** with Khalil Ammar (ICBA). Total Project Budget: $100,000.

2020: International Center for Biosaline Agriculture (ICBA), Adoption and Impact of Quinoa Among Farmers in Marginal Areas of the MENA Region. **Principal Investigator**. Grant Amount: $68,488.

2020: University of Wollongong in Dubai and ICBA, Resilient Agrifood Dynamism Through Evidence-Based Policies (READY). **Co-Principal Investigator** with Ioannis Manikas (University of Wollongong) and Khalil Ammar (ICBA). Total Project Budget: $1,500,000.

2019: IFAD and BADEA, Improving Agricultural Resilience to Salinity and Climate Change through Development of Pro-Poor Technologies and Management Strategies (RESADE). **Co-Principal Investigator** with R.K. Singh (ICBA). Total Project Budget: $7,700,000.

2019: International Center for Biosaline Agriculture (ICBA), EXPO Live Phase II: Value Chain Development for Innovative Halophyte-Based Food Systems. **Co-Principal Investigator** with Dionysia Angeliki Lyra (ICBA). Total Project Budget: $500,000.

Teaching Experience

2011–2015: Assistant Professor (teaching-focused), Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan *(held concurrently with the FLAG role, 2011–2012, and the DfID role, 2012–2015; 1–2 courses per term).*

Teaching responsibilities:

Agricultural Production Economics (AgEcEx731), undergraduate; spring terms, 2011–2015.

Agricultural Finance (AgEcEx836), undergraduate; fall terms, 2011–2015.

Introductory Statistics & Econometrics(AgEcEx630), undergraduate; spring terms, 2012–2014.

Research Methods & Projects (AgEcEx735; seminar-based teaching and student research supervision), undergraduate; spring & fall terms, 2011–2015.

Presentations and Talks

Invited Talks

2026: "Climate Shocks, Household Food Security, and Welfare in Afghanistan." Stanford University (FSE, Center for South Asia, Abbasi Program in Islamic Studies), April 22, 2026. Event | Slides | Video

2021: "Challenge and Opportunity of Quinoa for Food and Nutrition Security." Panellist with Warren T.K. Lee, chaired by Bo Zhou. FAO Regional Office for Asia and the Pacific (FAO-RAP), Bangkok, Thailand, November 7–8.

2021: "The Economics of Quinoa Production: Opportunity for Marginal Farmers." Panellist with Dr. José Graziano da Silva. Global Forum on Innovations for Marginal Environments (GFIME), Dubai, UAE, November 20–21.

Conference Presentations

2023: "Crop Diversification and Farm Household Welfare during Conflict: Afghanistan 2011–2017." 6th MaDE Workshop on Emerging and Developing Countries, Loughborough Business School (online), June 5–6, 2023. Slides

2021: "Assessing the Impact of Sand and Dust Storms on Agriculture." Food and Agriculture Organization (FAO), Bangkok (online), April 7–11, 2021. Slides

2020: "Salinization in Marginal Agricultural Areas: Implications for Global Food Security and Poverty." International Center for Biosaline Agriculture (ICBA), Dubai, UAE, June 27–28, 2020. Slides

2018: "Tests for Market Failures and Overcoming Transaction Costs in Afghanistan." Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, September 8–12, 2018. Slides

2017: "Status, Patterns, and Microeconomic Drivers of the Extent of Diversity in Crop Production: Evidence from Afghanistan." Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 15–20, 2017. Slides

2016: "Crop Diversification and Technical Efficiency: Evidence from Afghanistan." Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 12–16, 2016. Slides

Awards

2021: Co-recipient, Khalifa International Award for Date Palm and Agricultural Innovation (KIADPAI), awarded as Co-PI of the Project: "From Desert Farm to Fork: Value Chain Development for Innovative Halophyte-Based Food Products."

2015: PhD Scholarship Award, University of Nottingham and The Islamic Development Bank (IsDB).

2007: Graduate Scholarship Award, Purdue University and USAID-funded Afghan Merit Scholar Program (AMSP).

Service and Professional Activities

2019–present: Member, Agribusiness Sub-network, Association of Agricultural Research Institutions in the Near East and North Africa (AARINENA).

2019–present: Peer Reviewer for *The Journal of Development Studies*; *Food Policy*; *Sustainability* (MDPI); *PLOS ONE*; *Review of Development Economics*; *Humanities and Social Sciences Communications*; *Agriculture* (MDPI); *Cogent Economics and Finance*; *Journal of Social and Economic Development.*

Media and Outreach

2022: "How COVID-19 Crisis Undermined MENA States' Food Security." *Arab News*, November 20, 2022.

2021: "Agriculture in Marginal Lands." *Scholarly Community Encyclopedia*, September 2021.

2021: "New Study Reviews Status of Marginal Lands Worldwide." ICBA News Article, August 29, 2021.

2021: "World's Indigenous Peoples to Bear Brunt of Climate Change, Warns UN Paper." ICBA News Article, April 25, 2021.

2020: "ICBA's Quinoa Varieties Show Promise in Morocco." ICBA News Article, April 15, 2020.

Professional Development

2020: Responsible Conduct in Research: Research, Ethics, and Compliance. Collaborative Institutional Training Initiative (CITI). Completed online, March 24–April 4.

2013: Leadership and Governance of Development Programs and Projects. Crown Agents International. Dubai, United Arab Emirates, November 11–15.

2012: Management of Agriculture and Rural Development Projects. Asian Institute of Technology (AIT). Bangkok, Thailand, November 21–December 1.

2008: Academic English and Cross-Cultural Training Program. International Centre for Intercultural Communication (ICIC). IUPUI, Indianapolis, IN, USA, October 29, 2007–May 30, 2008.

Research Methods and Skills

- **Econometrics & Causal Inference:** Applied econometrics; causal inference; quasi-experimental methods (IV, DiD, RDD, matching); panel-data methods; discrete-choice models; experimental and field-experimental methods; welfare and distributional analysis.
- **Agricultural & Field Research:** Farm and household surveys; agricultural production and farm economics; farmer adoption and technology uptake; agricultural productivity; value-chain analysis; rural livelihoods; food security; field-based data collection and research design.
- **Data & Empirical Research:** Survey sampling and design; administrative, climate, longitudinal, and geospatial data; dataset construction and harmonization; quantitative research design; mixed-methods research; reproducible data and analysis workflows.
- **Research Software & Data Tools:** Stata, R, Python (Pandas, NumPy, Statsmodels, Seaborn, Matplotlib), QGIS, Survey Solutions, Qualtrics, Excel.
- **Research Management:** Grant writing and proposal development; multidisciplinary research collaboration; project management; research supervision and mentoring; technical and scientific reporting.

Languages

- English (fluent)
- Pashto (native)
- Dari/Persian (fluent)
- Arabic (basic)